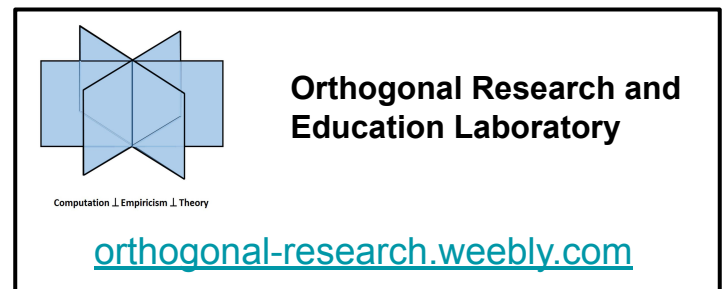
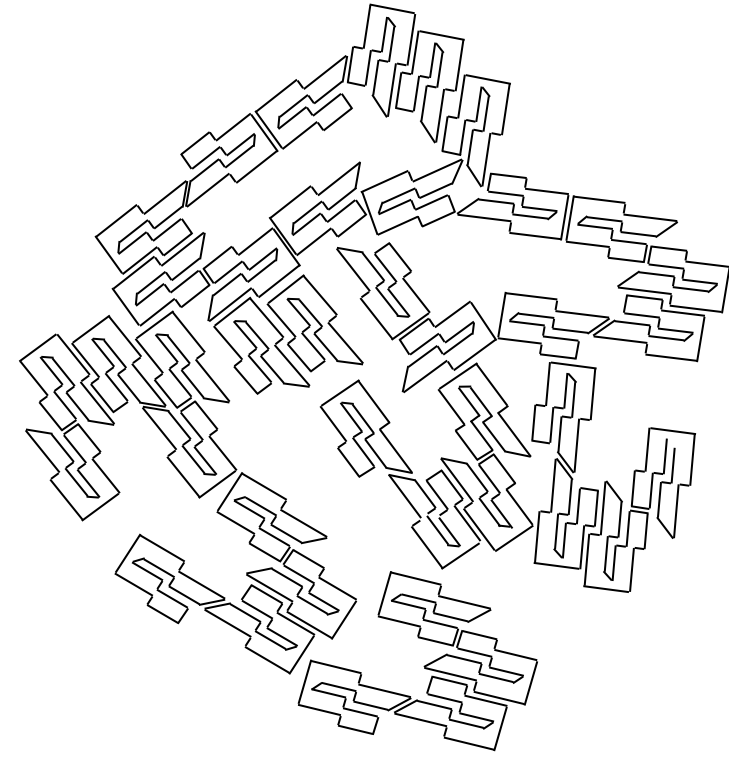


Phylogenetically-Mixed Architectures

The development and evolution of agentive complexity

P26, fp192



Phylogenetic Mixing as “In-silico Hybridogenesis”



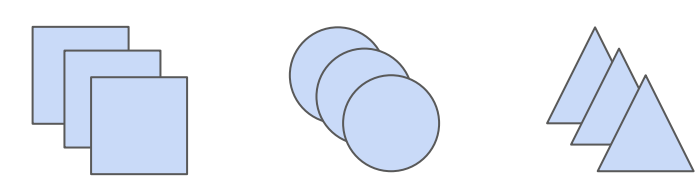
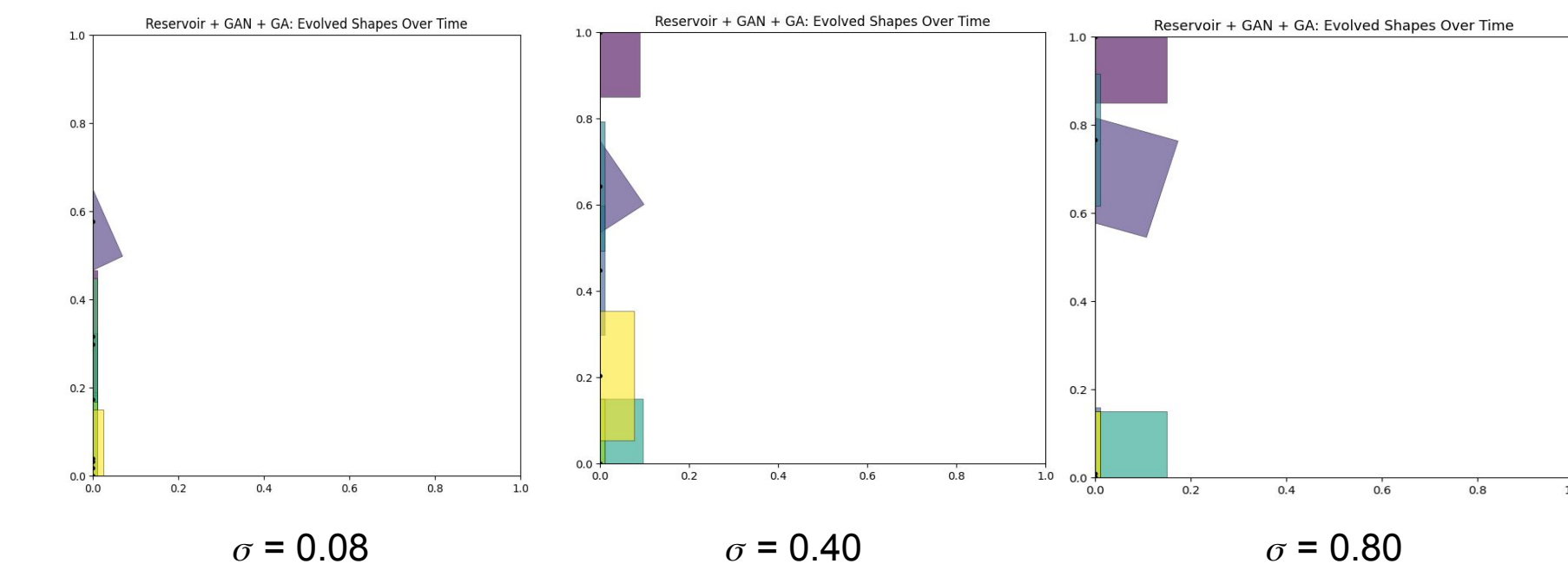
Different nervous system circuits = composite neural architecture.

Place inside of a generic body, or a hybrid body (with features of the circuit in question).

Objective 1: evolve from a common ancestor (simple motif)

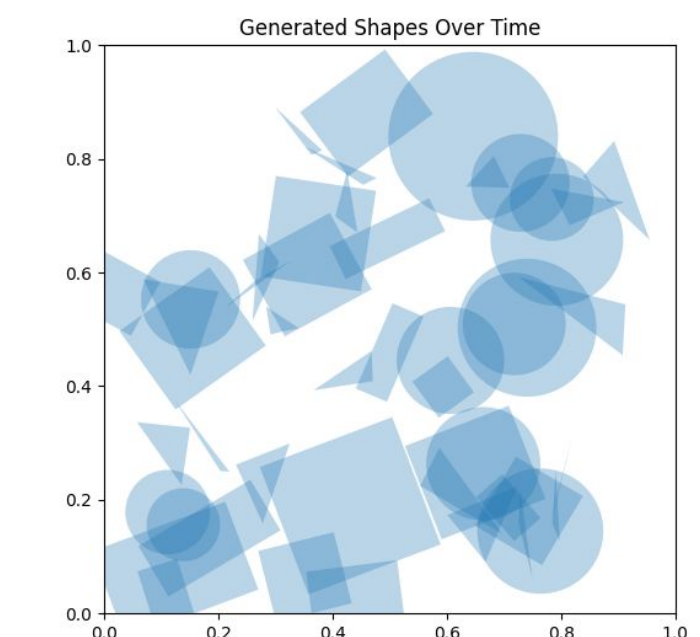
Objective 2: stitch phylogenetically-distant motifs together.

Noise as the Output of Reservoir Network

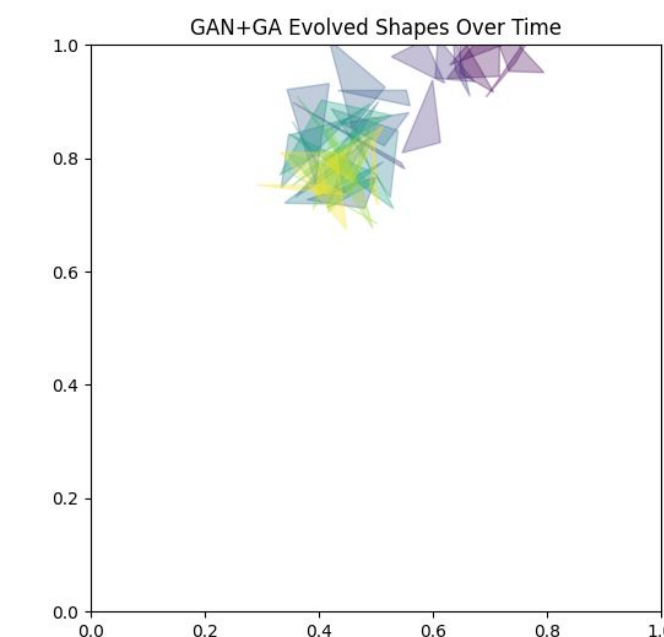


Start with a population of generated shapes, then select the most viable.

Generated by GAN from Reservoir Network output



Generated by GAN + GA (evolutionary process)



Top-down generation of embodied agent forms

Morphological Computation:

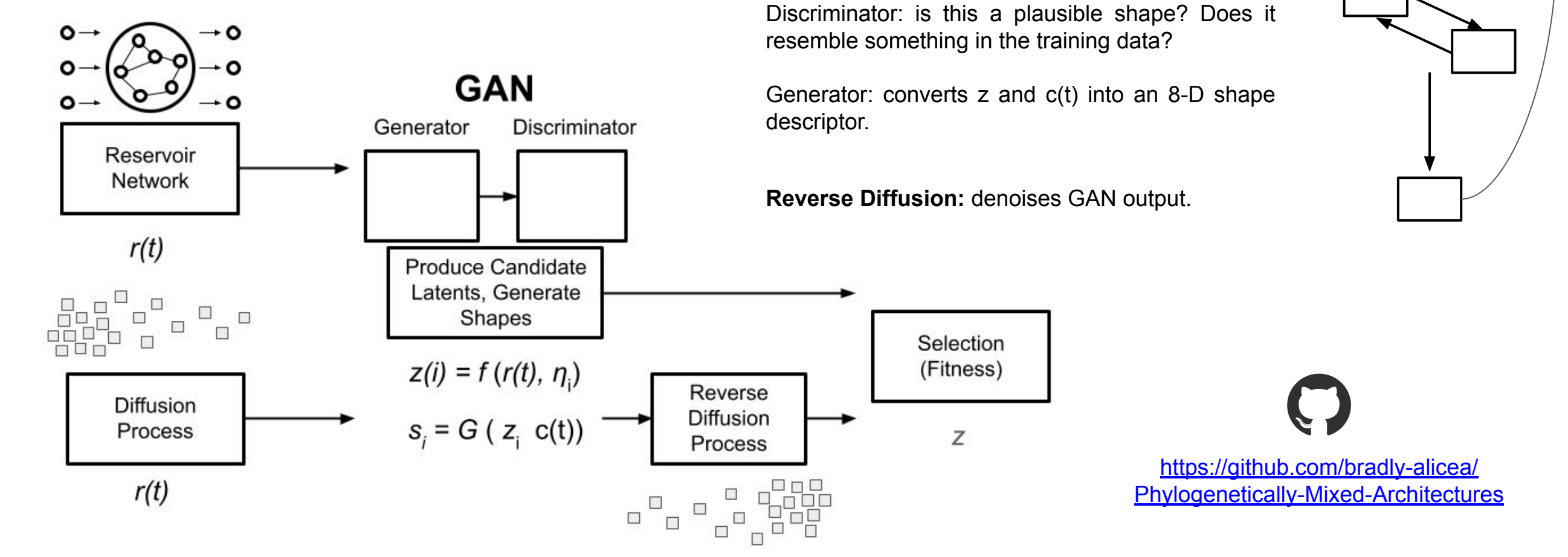
- Cheney (2017): design pipeline that establishes an optimization mechanism for specialized and optimized complex morphologies.
- D'Angelo et.al (2026): an embodied agent design pipeline is an important component of benchmarking agent performance.

Diffusion Process: produces a high-dimensional state $x(t)$, project to a context vector $c(t)$

η is the noise-strength parameter: random signal is injected into the forward diffusion process.

Phylogenetic mixing (in-silico hybridogenesis):

- provides a means to develop novel and consensus forms while also assembling more complex morphologies from disparate components.

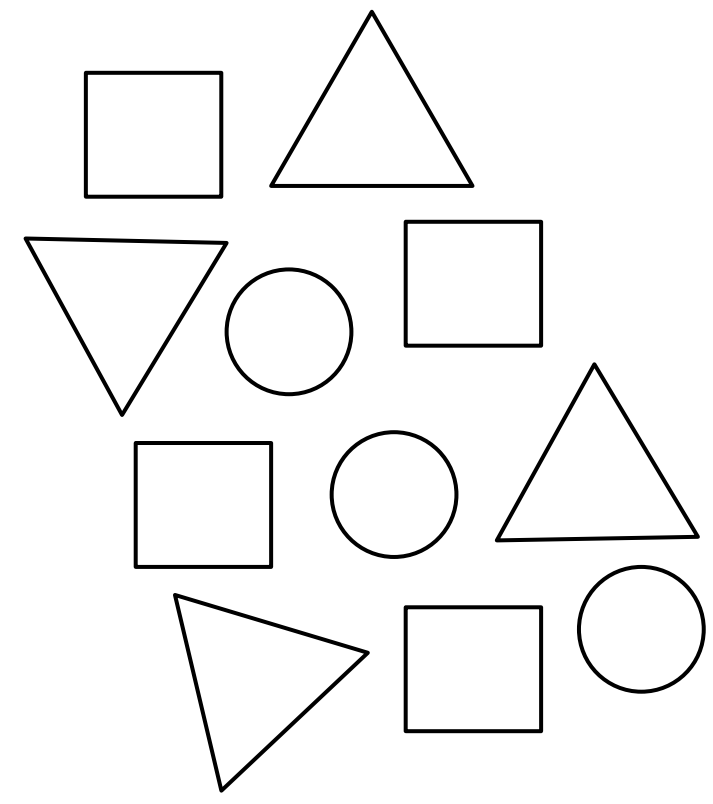


<https://github.com/bradly-alicea/Phylogenetically-Mixed-Architectures>

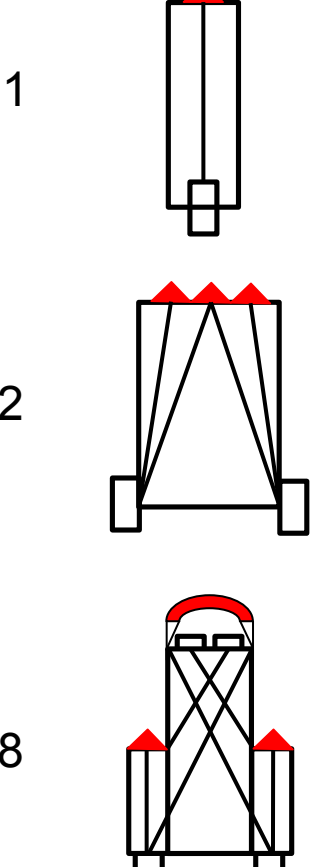
Shapes Dataset

Sample and generate ensembles of body parts (left).

Selection is used to explore the possibility space.



Braitenberg Vehicles Dataset

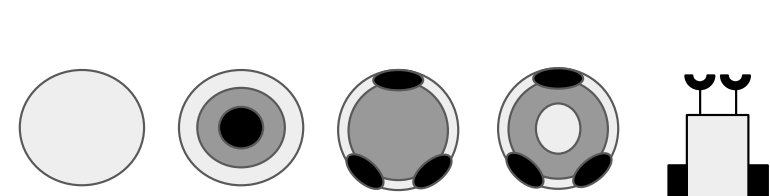


Sample and generate morphologies of vehicle bodies (left).

Sample and generate descriptions of behaviors and associated connectivity patterns (right).

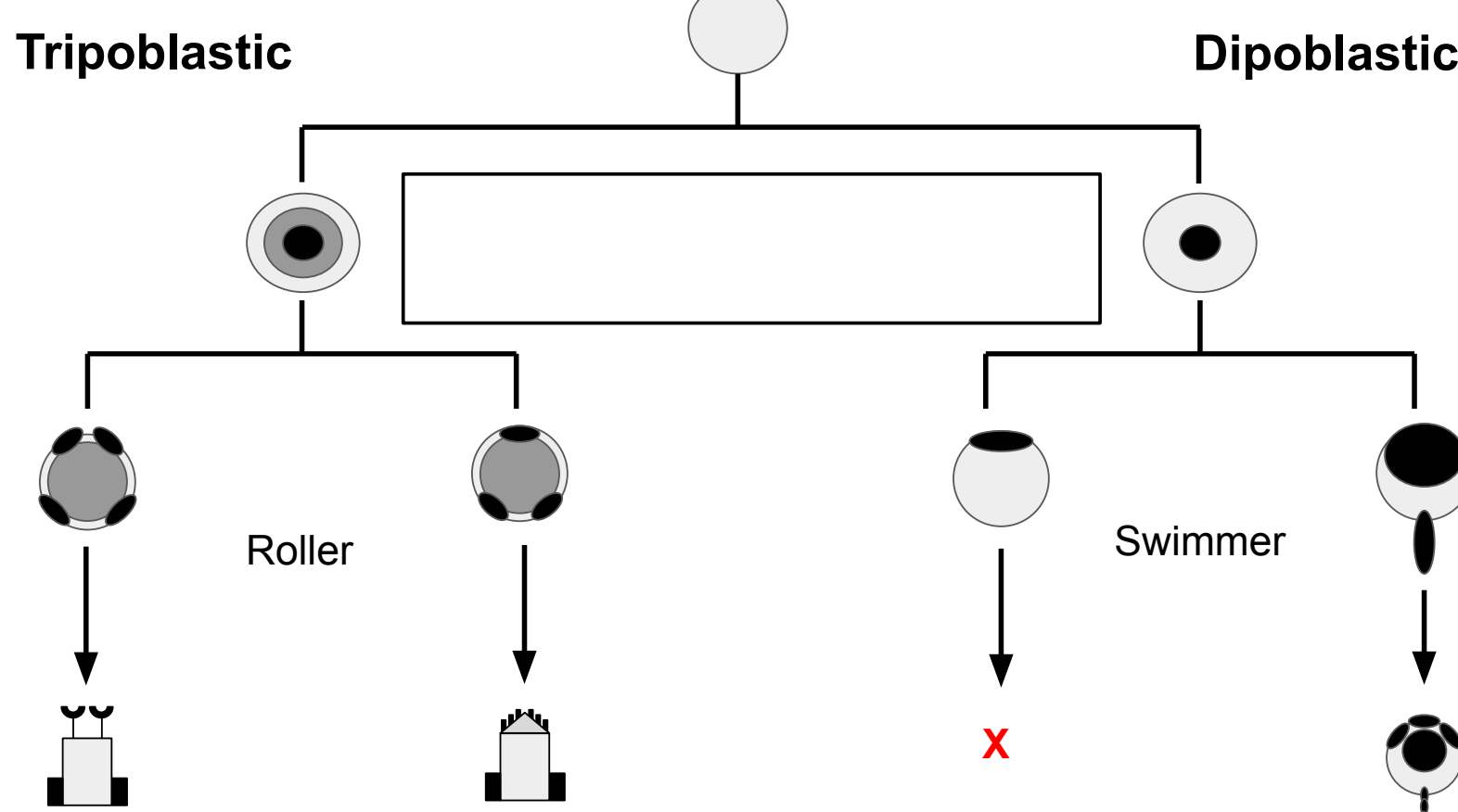
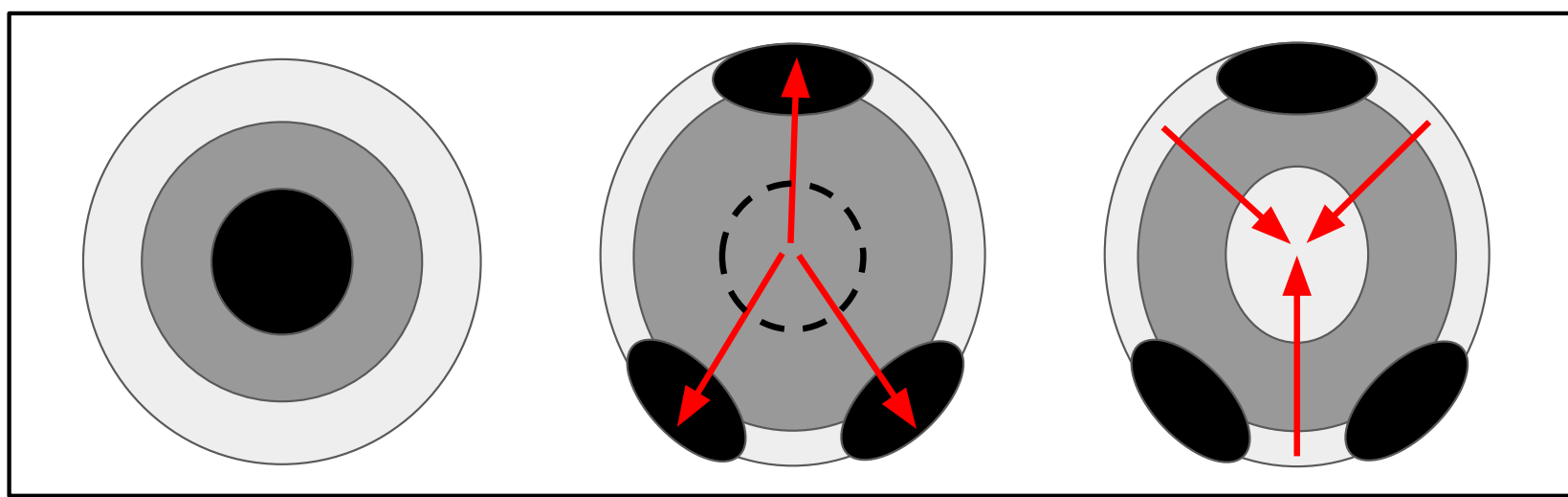
Description	Theme	W	S	Cs
1. Getting Around	Locomotion	U	F	2
2a. Near/Aggression	Trope/Action	C	F	2
2b. Near/Aggression	Trope/Action	U	E	2
3a. Love/Liking	Trope/Action	U	E	2
3b. Love/Liking	Trope/Action	C	E	2
4. Values and	Preferences			3
5. Logic	Logic			3
6. Selection	Selection			3
7. Concept	Concepts			4
8. Space, Things, and Movement	Spatial and Movement			4
9. Shapes	Perception			4
10. Getting Ideas	Ideas			5
11. Rules and Regulations	Rules			5
12. Traits of Things	Things			5
13. Foresight	Foresight			6
14. Equanim and Patience	Equanim and Patience			6

As seen at



Triploblast Model of Embodiment

Agent embryogenesis proceeds as a series of morphogenetic movements (acquisitions).

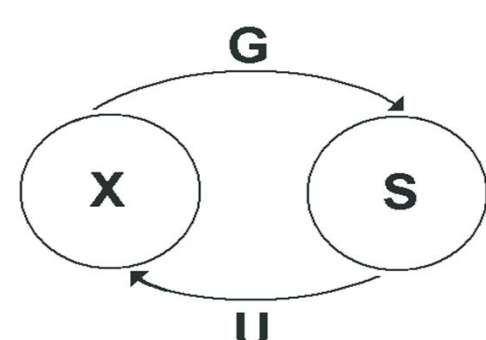


Computational version of this generative process is needed

Why Agentic Coding?

Agentic coding can serve as an iterative research sketchpad.

- Provides a means to rapid prototype models.
- Integrated development via refined prompts.
- Extract datasets from text and RAGs.



Intelligence Augmentation (*sensu* Ashby)

S is the Amplifier (LLM), X is the user (Human)

Human

What models are most plausible for developing embodied phenotypes?

How does this fit into a pipeline? What do the details mean? I'll suggest what data to benchmark models with.

I'll test the code and evaluate both the codebase and output.

I'll suggest ways to visualize the results using Matplotlib.

Machine

LLM generates code (steps in analysis, toolbox handling) for implementing candidate models.

LLM refines definition and technical details, helps to create benchmark datasets.

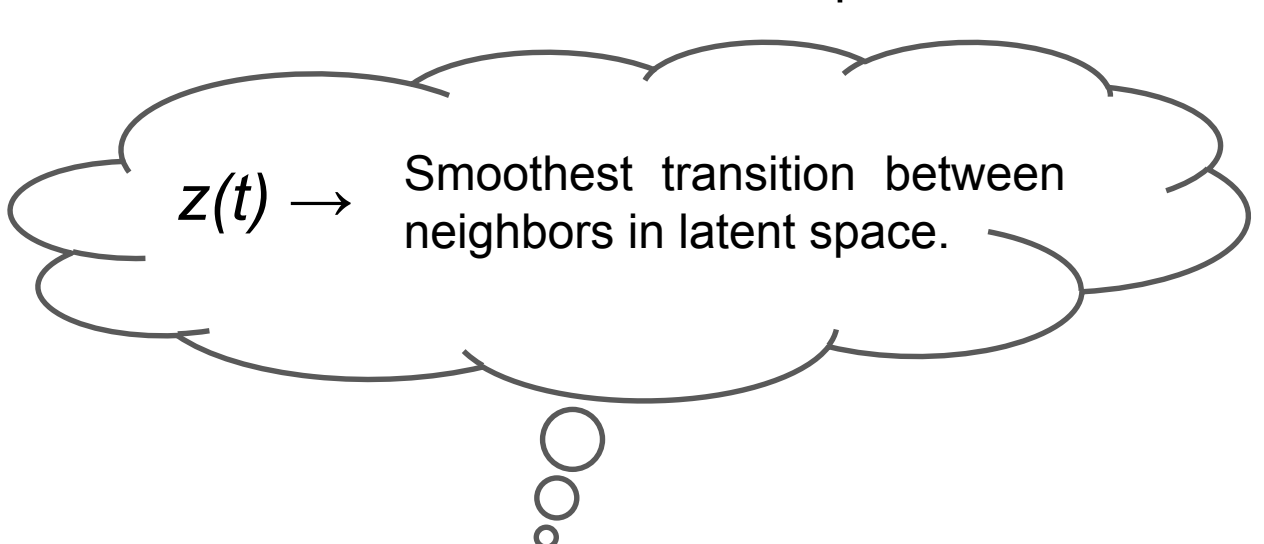
LLM refines parts of the code and dataset that do not make sense or do not execute.

Genetic Algorithm

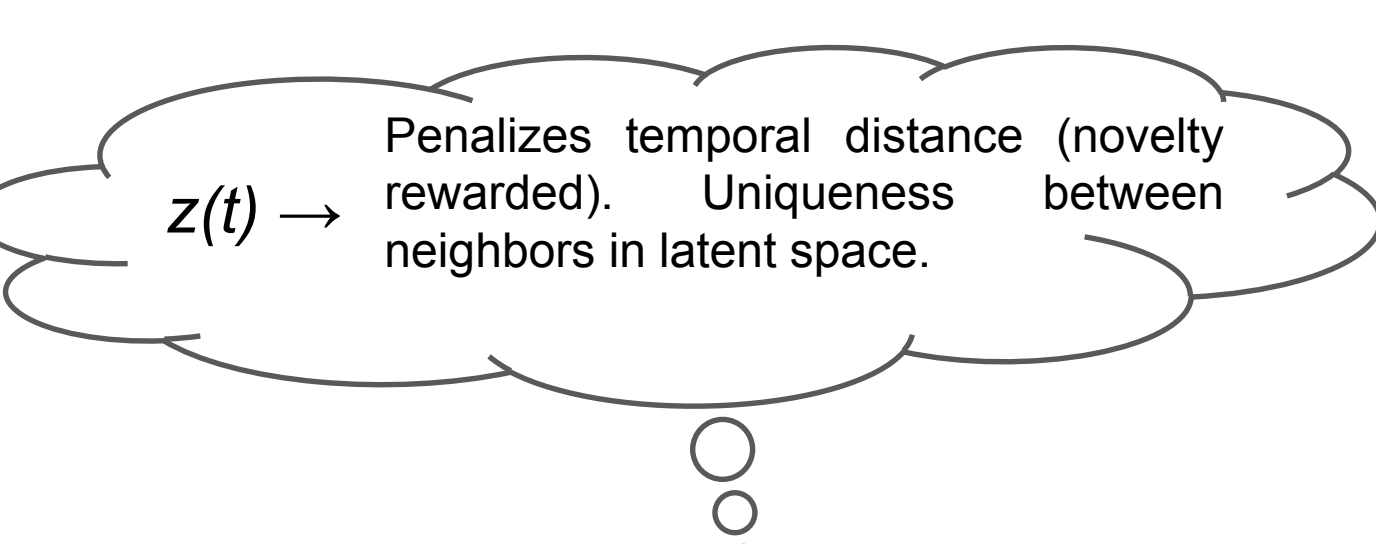
Crossover	$\alpha = \text{torch.tensor}(\text{rng.random}(), \text{dtype}=\text{torch.float32}, \text{device}=\text{device});$ $\text{child} = \alpha * p1 + (1 - \alpha) * p2$
Mutation	$\text{mut} = \text{torch.randn_like}(\text{child}) * (0.05 + 0.02 * \text{mg.random}());$ $\text{child} = \text{child} + \text{mut}$

GAN Output

$z(t) \rightarrow \text{fitness function} \rightarrow \text{selection}$

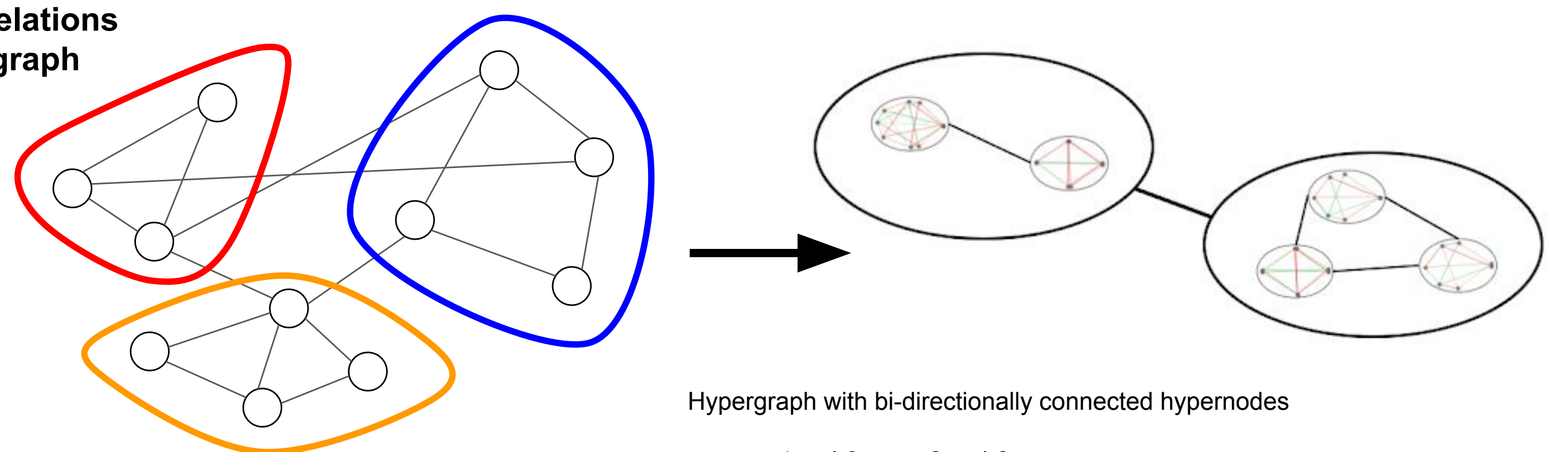


Smooth Descent with Modification



Creativity

Multi-scale Relations in a Hypergraph

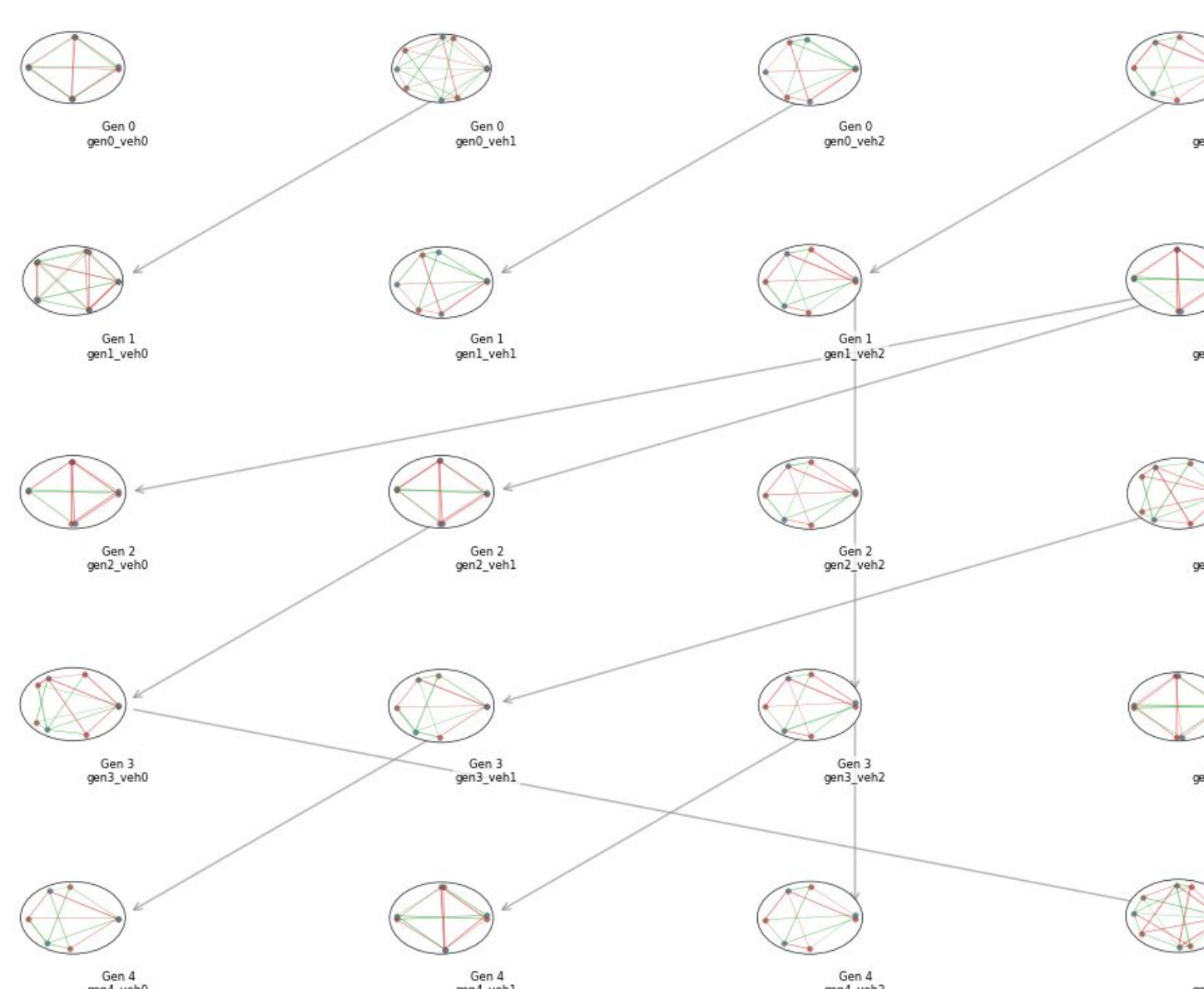


Hypergraph with bi-directionally connected hypernodes

- gen4_veh3 + gen2_veh0.
- gen4_veh0 + gen4_veh1 + gen4_veh2.

A phylogenetic model of BVs that can be used to build a hypergraph

- various emerging lineages of vehicle morphology.



How do we make this pipeline more ecologically friendly?

Lowering carbon emissions for the pipeline.

Will running locally with open-source models (rather than frontier models via API) help?

What about using a RAG to more precisely define the training set (not shapes, often will be 3-D)?

Strategies for carbon prompting: use as few prompts as possible to generate a sparse representation. Once we refine the pipeline, minimize the number of prompts used.

Code and other Materials on Github

<https://github.com/bradly-alicea/Phylogenetically-Mixed-Architectures>

GAN + GA evolved steps over time

Novelty-first Dendrogram (Hierarchical Clustering based on prior shape history)